



A teaching-free welding method based on laser visual sensing system in robotic GMAW

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Abstract

At present, the majority of welding robots belong to the teach-and-playback category in welding manufacturing engineering applications; trajectory teaching in advance of welding is time-consuming and lack of efficiency. The currently published welding seam tracking methods are also based on existing trajectories. Intelligent Welding Manufacturing (IWM) is the current research focus in welding manufacturing. The core technology of IWM is Intelligent Robot Welding Technology (IRWT). A sensing-technology-based system was the key to realize IRWT. In this paper, a teaching-free welding method based on laser visual sensing system (LVSS) for robotic gas metal arc welding (GMAW) is studied. First, a LVSS was established. A fast-unified calibration method for LVSS is proposed to improve the calibration efficiency. Then, using an image processing method based on prior knowledge, feature point with sub-pixel accuracy can be obtained in real-time. Finally, an online welding trajectory planning method is proposed to implement teaching-free welding. In order to verify the accuracy and robustness of the proposed method, experiments on V-grooves and fillet welds were performed. The results showed that the control accuracy on the V-groove and the fillet welds is suitable for most robot welding applications.

Keywords Intelligent welding · Teaching-free · Real-time online planning · Laser visual sensing system · Image processing

1 Introduction

After undergoing manual manufacturing, mechanized manufacturing, and automated manufacturing, intelligent manufacturing has become the focus and research hotspot of manufacturing research. IWM is the future development trend of intelligent welding manufacturing. Robot welding plays a vital role in modern manufacturing; IRWT is the core of IWM [1–6].

Most welding robots currently in service are still teach-and-playback category, requiring manual teaching and welding trajectory point programming. The accuracy of teaching depends on eye observation and operators' experience. Under conditions of a wire with a diameter of 1.2 mm, it is generally

considered acceptable that the error does not exceed 0.5 mm. Although it is a great improvement over manual welding, the teaching step is inefficient and requires a high degree of concentration. In addition, the teach-and-playback robot cannot recognize changes in the welding environment, such as assembly errors, thermal deformation, and errors in the pre-machining process, which may result in unstable welding quality or even welding failure.

Today, most industrial robot brands have their own offline programming software, such as ROBOGUIDE from FANUC, RobotStudio from ABB, and Simi Pro from KUKA. The commercial offline programming method can do the job without considering the thermal deformation during the welding process and the deviation of the initial position. Offline planning is also a hot topic in research field. Larkin et al. [7] proposed an automated offline programming method for welding robots, which inputs a 3D model of the weldment, and outputs a robot program to execute the job. Erdős et al. [8] developed a toolbox for remote laser welding, which solved not only the trajectory planning problem but also the welding task, path planning, workpiece placement, inverse kinematics calculation, and generating of robotic program code. Zhang et al. [9] solved the complex curve joint planning problem of the dual

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robot system. The simulation results showed that the joint displacement curve is smooth and continuous, and no joint velocities exceed the working scope. Liu et al. [10] presented a curving fitting method for intersection curve welding, which verified the correctness and flexibility requirement. By dragging the workpiece model into either the commercial version or laboratory version of the software, then output the robot executable program that arranges the execution trajectory, offline program research greatly improves the flexibility and efficiency of the robot trajectory planning. The research focus on offline programming of robots is the complexity and flexibility of the trajectory, rather than the environmental changes and perceptions in real-time processes.

The focus of the offline programming method is to solve the current path performed by the robot, which is the same as manual teaching, without considering the dynamic deformation during thermal process of welding. There would be a deviation between the real trajectory and the planned one. With the help of sensing technology, the welding robot can detect the deviation and correct it, known as weld seam tracking. Generally, weld seam tracking is established by extracting the deviation between the current torch position and the center of the weld seam, and then correct it through a closed-loop control method. Weld seam tracking can be achieved with different types of sensors, such as rotating arc sensors [11, 12], acoustic sensors [13, 14], spectral sensors [15, 16], and vision sensors. Among all the above sensing methods, visual sensing has attracted the most attention due to its non-contact measurement and rich information acquisition. Shao et al. [17] performed weld seam tracking with passive vision sensor, particle filter, and Hough transform which were used to increase the robustness. Nilson et al. [18] established a camera-based system solving tracking problem for closed-square-butt joint. Hough transform and Kalman filter were used to estimate the robot offset; offline evaluation of the method proposed shows a promising result.

The visual sensing is divided into active mode and passive mode through the additional light sources. Passive vision uses arc as the light source; images of the weld pool area are obtained through a suitable dimming filter lens. A novel visual sensing system [19] was established, and an image acquisition and processing method for robot GMAW seam tracking were proposed [20] by Xu et al. The results showed that it can achieve the tracking accuracy requirement of most welding applications. Shen et al. [21] solved the penetration control problem in GTAW by molten pool image processing and PID controller. He et al. [22] focused on weld seam profile extraction with visual attention and EWMA method using active vision sensor; the result shows that the method lays a good foundation for general weld seam profile extraction and a potential applications prospect. Du et al. [23] improved the robustness of image processing algorithm by introducing a CNN model

in the GMAW tracking process, especially when the images with strong interferences. The result shows the proposed method is effective and accurate.

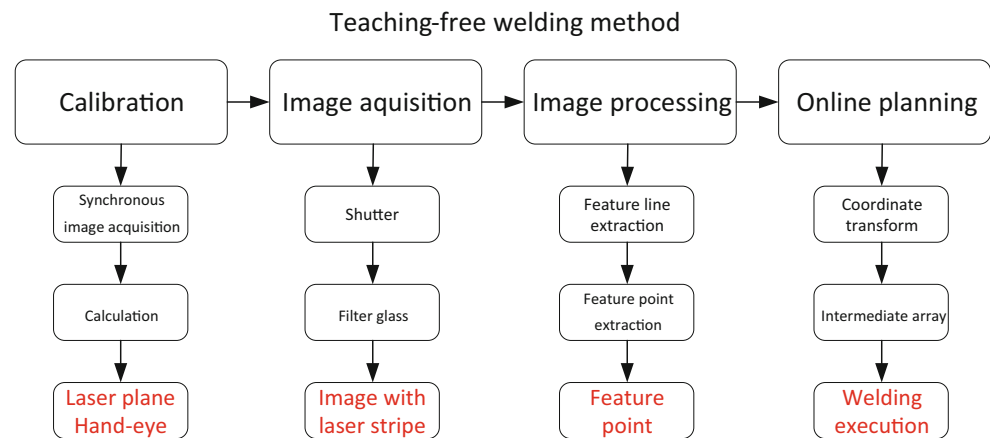
The study of active vision sensing, especially structured light vision, is a popular topic. Due to the fact that structured light vision does not directly observe the molten pool area, the optical system is greatly simplified, making it a very reliable information efficiency. Xu et al. [24] used a circular laser sensor to achieve weld tracking in GTAW. Luo et al. [25] reported a laser vision method for weld seam tracking of titanium alloy welding with an accuracy of less than 0.4 mm. Fang et al. [26] proposed a laser-based self-tuning fuzzy controller for fillet weld tracking, extracting deviations and controlling an external axe to eliminate deviations. Zou et al. [27, 28] used the morphological image processing and CCOT algorithm to achieve seam tracking. The results show that the measurement accuracy can reach $\pm 0.2\text{mm}$ at 50 Hz. According to the reported research, the proposed weld seam tracking methods require an initial weld trajectory as a comparison and deviations extracted from the image as an adjustment amount. The robot welding process still needs to be taught in advance.

Offline programming cannot adapt to the thermal deformation during the welding heating process. Passive visual sensing is sensitive to optical parameters and lacks robustness when welding interference is strong. External axes are required to adjust deviations according to reports on active visual sensing-based weld seam tracking methods. In addition, the deviation value is still a relative value, and the adjustment movement cannot match the welding movement itself. Therefore, the real-time online planning method based on active vision is the best choice for intelligent welding teaching problems.

This paper studies the teaching-free intelligent welding method of robot GMAW. The second part introduces a teaching-free robotic intelligent welding method, including intelligent robot welding system using LVSS, fast-uni-axial calibration method for LVSS, improved image processing method, and welding trajectory online planning method based on intermediate array. In the third section, experiments were carried out on V-groove and fillet welds to verify the accuracy, real-time performance and robustness of the proposed method. The axial error of the welding trajectory was about 0.2mm. Some conclusions are drawn in the fourth section.

2 Methodology

There are two key issues in implementing teaching-free welding. One is the precise extraction of the trajectory points and the other is how to proceed continuously during the welding. The improved image processing method proposed in this paper can extract the weld feature points

Fig. 1 Flow chart of teaching-free welding method

as welding trajectory points in real-time under the strong interference of welding with subpixel accuracy. By storing the planned but unexecuted trajectory points in the intermediate array, and the continuous extraction of feature points during the welding process, the trajectory is executed continuously by the robot.

Figure 1 shows the flow chart of the entire process of the teaching-free welding method. The method can be divided into four main parts: calibration, image acquisition, image processing, and online planning.

2.1 Construction of robot welding experiment system

In order to realize the robot welding without teaching, an intelligent system based on the robot welding station was established. Figure 2 shows the structure of the system. The basic robot welding station consists of industrial robot, welding power, torch, wire feeder, and protection gas cylinder. The LVSS developed in this paper includes a laser vision sensor and an information processing center (IPC). IPC fully controls the motion of the laser vision sensor, communicates with the robot control cabinet via Ethernet, reads/writes the

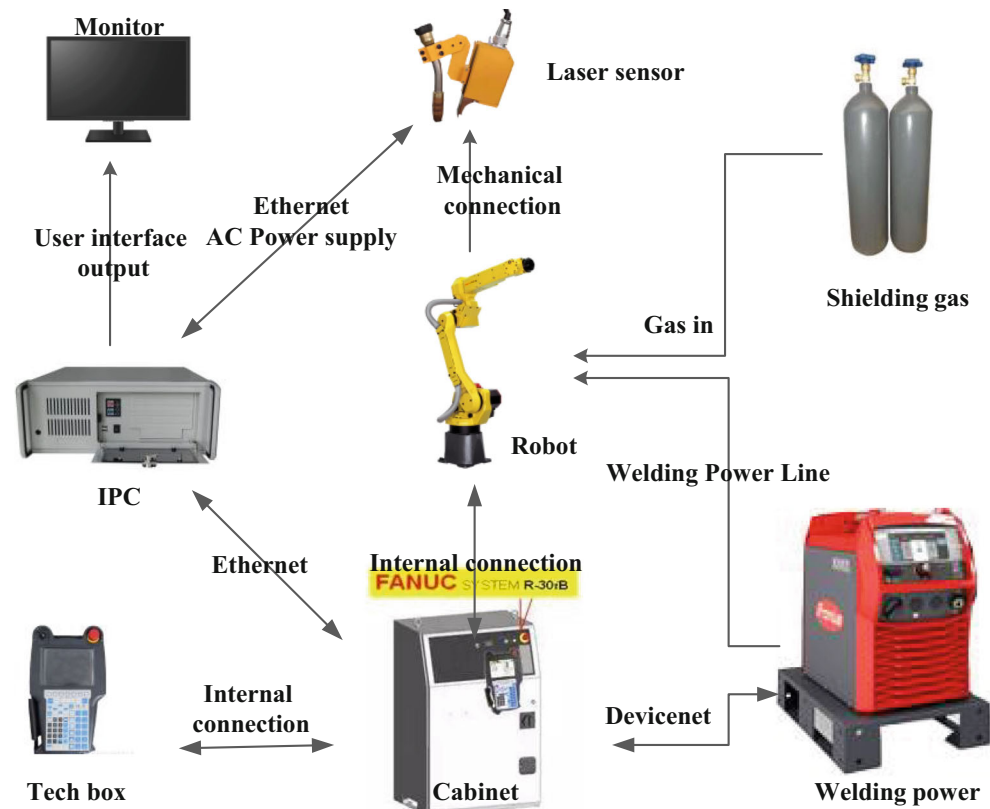
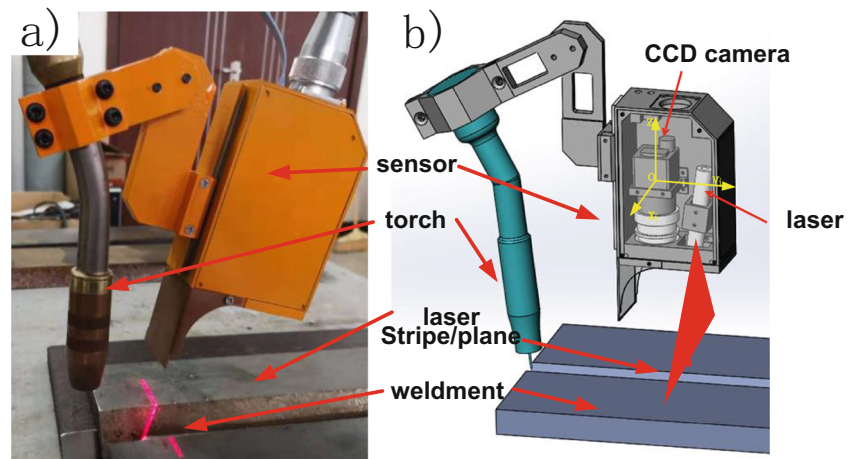
Fig. 2 Robotic welding system based on LVSS

Fig. 3 Laser vision sensor

robot status, and continuously acquires target point of robot movement.

Sensors play a vital role in robotic intelligent welding and are the medium for communicating robot system and the environment. The robot welding system equipped with sensors has the ability to recognize environmental information. As shown in Fig. 3, this paper designs and develops a laser vision sensor. The CCD camera, laser generator, filter glass, and dust cover are installed inside the laser vision sensor.

In this paper, Fanuc M-10iA robot and the Fanuc R-30ib cabinet with system version 7.7 are used, and welding power is provided by the Fronius TPS 400i. The parameters of the camera and laser generator in the laser vision sensor were listed in Table 1.

Since the high requirements to implement real-time control of the welding process, a software program was developed with C++. The software includes an image acquisition module, an image processing module, and a communication module. The flow chart of the software which is shown in Fig. 4 a and b is the user interface of the software.

Table 1 Parameters of laser vision sensor

Camera (MER-200-14GM)	
Resolution	1628(H) × 1236(V)
Frame rate	14 Hz
Sensor type	1/1.8" Sony ICX274 AL/AQ CCD
Pixel size	4.4 μm × 4.4 μm
Laser generator	
Power	200 mw
Line width	0.8 mm
Wave length	660 nm

2.2 Laser vision model

In passive vision sensing, there are two calibration methods. The first one is in the case of gas-shielded tungsten arc welding (GTAW) as shown in Fig. 5, the posture of the welding torch is unchanged, and the shooting area is relatively fixed. It is only necessary to calibrate the scale factor k of the corresponding image area [29]. Then, the corresponding features in the image can be calculated by formula (1).

$$\begin{aligned} k_v V &= X_{real} \\ k_h h &= Y_{real} \end{aligned} \quad (1)$$

where k_v and k_h present the scale factor in horizontal and vertical direction in image coordinate, and v and h are the numbers of pixels in horizontal and vertical direction in image coordinate respectively.

The advantage is that the amount of calculation is small, and the obtained deviation value is a relative value between the real value and the planned value, which can be directly used to correct the welding trajectory of the weld tracking based on passive vision. The disadvantage is that it does not adapt to changes in posture or space welding.

The other calibration method for passive vision is to realize stereoscopic vision through multi-position-multi-posture vision using Zhang's method [30] to calibrate the hand-eye relationship in formula (2), which was used to convert image coordinate (eye) to spatial coordinates (hand).

$$s\tilde{m} = A[R, t]\tilde{M} \quad \text{with } A = \begin{bmatrix} \alpha & c & \mu_0 \\ 0 & \beta & v_1 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

where s is an arbitrary scale factor, $[R, t]$, called the extrinsic parameters matrix, is the rotation and translation that associates the world coordinate system with the camera coordinate system. A is called intrinsic parameters matrix. The point $(u_0, v_0)_{\text{image}}$ is the principle point of the image. α and β are the

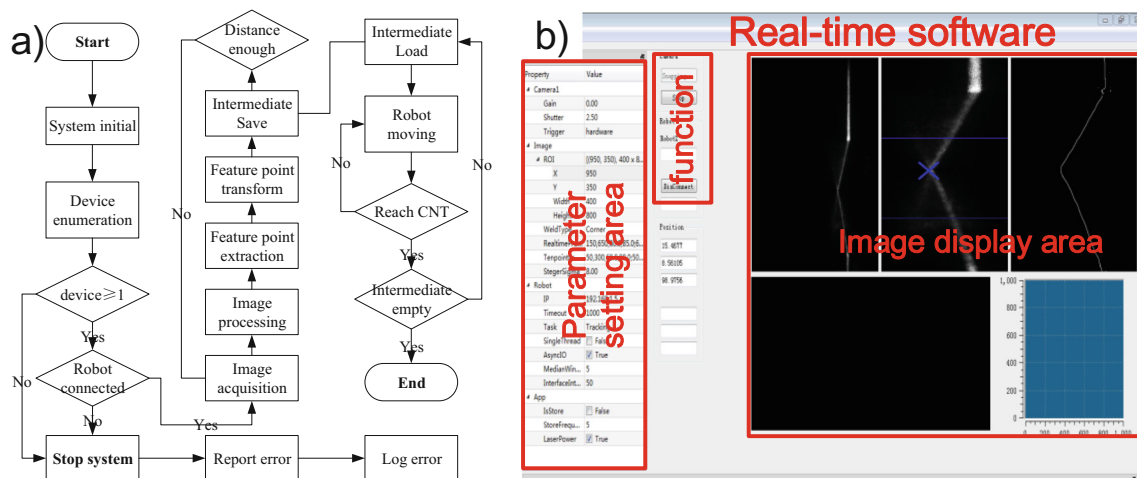


Fig. 4 Flow chart and interface of real-time welding control software

scale factors in u and v axis of image coordinate, and c is the parameter describing the skewness of the two axis. $\tilde{m}$ and $\tilde{M}$ are the augmented vectors of the coordinates in image coordinate system m and world coordinate system M .

Absolute coordinates can be obtained directly, which can be used for trajectory planning, but the field of view of passive vision is not suitable for real-time applications.

In visual sensing, when a set of points in a 3-D space is mapped into a 2-D image, the depth information of points in the image will be lost, which means that in passive vision, the target of the acquisition is the 2-D set of points; only 2-D information of the points of 2-D set is obtained by a single frame of image. As shown in Fig. 6, each point in line Aa will be projected to position a in the image. The stereoscopic image acquires supplementary information of the corresponding space by additional shooting. In order to determine the 3-D information of the 2-D point set, supplementary shooting and image matching must be performed.

While in structured light vision, the collected target is a 1-D point set. The light plane in the camera coordinate is relatively fixed, and the points in the 1-D point set all come from the structured light plane; the structured light plane equation complements the hand-eye relationship, so a single frame image can get the complete 3-D information of the 1-D point set.

2.3 Fast-uniated calibration for LVSS

The calibration process is very important for the accuracy of the vision system, and its efficiency is especially important for its application. The calibration process for LVSS was divided into three separate sections: tool center point (TCP) calibration, laser plane calibration, and hand-eye relationship calibration, as shown in Fig. 7. The purpose of calibration is to determine the conversion relationship from the image coordinate system to the robot world coordinate system.

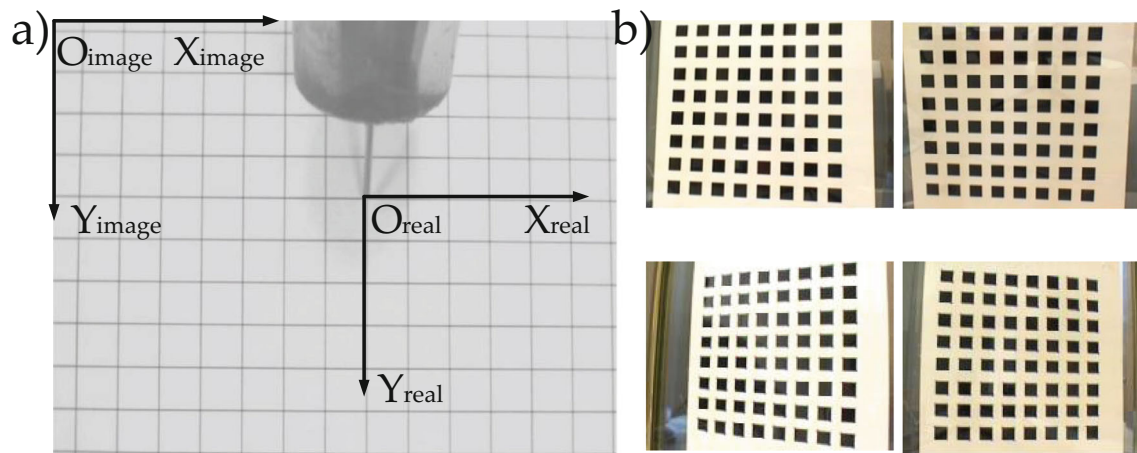


Fig. 5 Passive vision calibration **a** scale factor and **b** hand-eye relation calibration

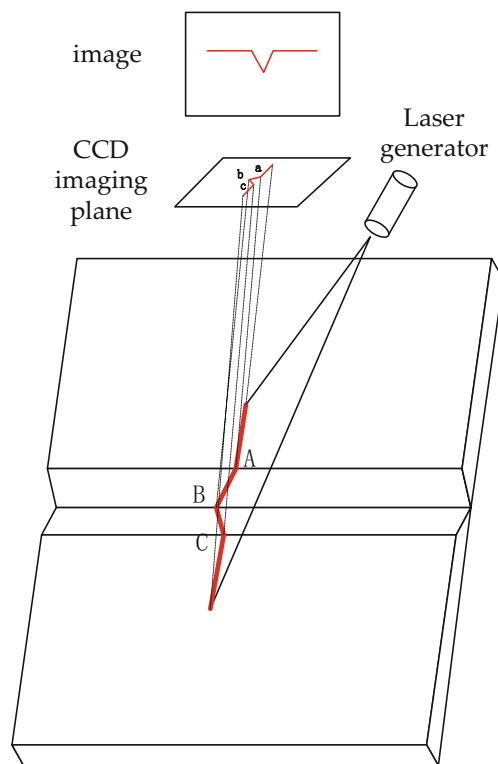


Fig. 6 The working principle of LVSS

TCP calibration is the first step, which can be carried out with 3-point method or 6-point method according to different stick out that weld process requirement.

Laser plane calibration is to determine the laser plane function in the camera coordinate system. The laser plane calibration needs to acquire images at different positions and postures; the laser plane function can be expressed as formula (3):

$$Ax + By + Cz + D = 0 \quad (3)$$

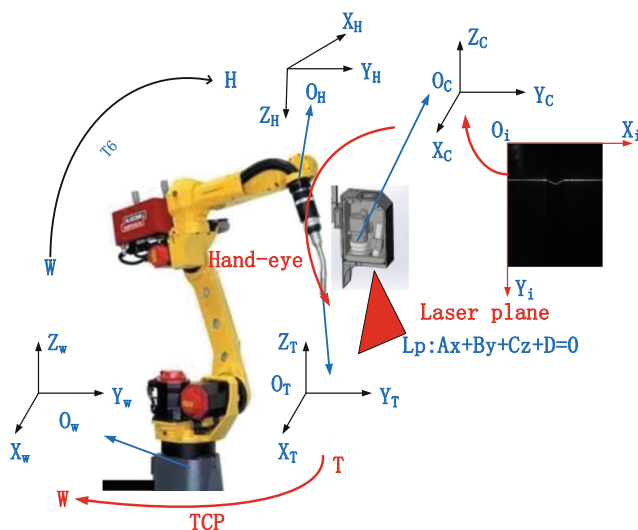


Fig. 7 Coordinate system of LVSS

The hand-eye relationship calibration is to determine the relationship between TCP and camera; Zhang's method is the most widely used to resolve hand-eye relationship H expressed as formula (4), which also needs images acquired at different positions and posture.

$$H = A \begin{bmatrix} R & T \\ 0^T & 1 \end{bmatrix} \quad (4)$$

where A is the camera intrinsic parameters, and $R_{3 \times 3}$ and $T_{3 \times 1}$ are the rotation and translation matrix respectively.

Laser plane calibration and hand-eye relationship calibration constitute the calibration of LVSS. Previously, these two steps were performed independently, and it was necessary to repeat the operation of changing the position and posture of the robot. Both laser plane and hand-eye relationship calibration require the images acquired at different robot positions and postures, but the shooting requirements for image acquisition are different. Laser plane calibration requires an image with only laser stripe, while the markers on the calibration board are useful for the hand-eye relationship calibration.

In order to improve the efficiency of the calibration process, this paper proposes a fast-united calibration method. Figure 8 shows the flow chart and interface of its implementation software. The proposed method combined laser plane calibration and hand-eye relationship calibration together.

Figure 9 shows the flow chart of the fast-united calibration method. The key is synchronous image acquisition. In the same robot position and posture, two images can be obtained by one image capture trigger, as shown in Fig. 10

According to the requirements of image acquisition, Short Shutter (2 ± 0.5 ms) With Laser (SSWL) images are used to calibrate the laser plane function, while the hand-eye relationship calibration requires Long Shutter (230 ± 30 ms) No Laser (LSNL) images. During the calibration process, the calibration software will capture the image and records the position of robot during the acquisition. After collecting enough data (image and position), a MATLAB program will be called to calculate the calibration results: the laser plane function and the hand-eye relationship matrix. The laser plane function and the intrinsic parameters of camera will be applied in a real-time software to convert feature point from image coordinates $m = (h, v)_{\text{image}}$ to camera coordinates $c = (x, y, z)_{\text{camera}}$ and prepare to respond to the robot's request. The hand-eye relationship matrix is combined with the TCP calibration result to obtain the sensor frame $S_f = (x, y, z, w, p, r)$, which is used to convert the coordinates in the camera coordinate to a Cartesian coordinates, whose original point is at the end of the torch; axes are parallel to the robot world coordinate. When the robot performs online planning, it will continuously send requests to the IPC. The IPC responds the request with feature points information in the camera coordinate form I , then transferred

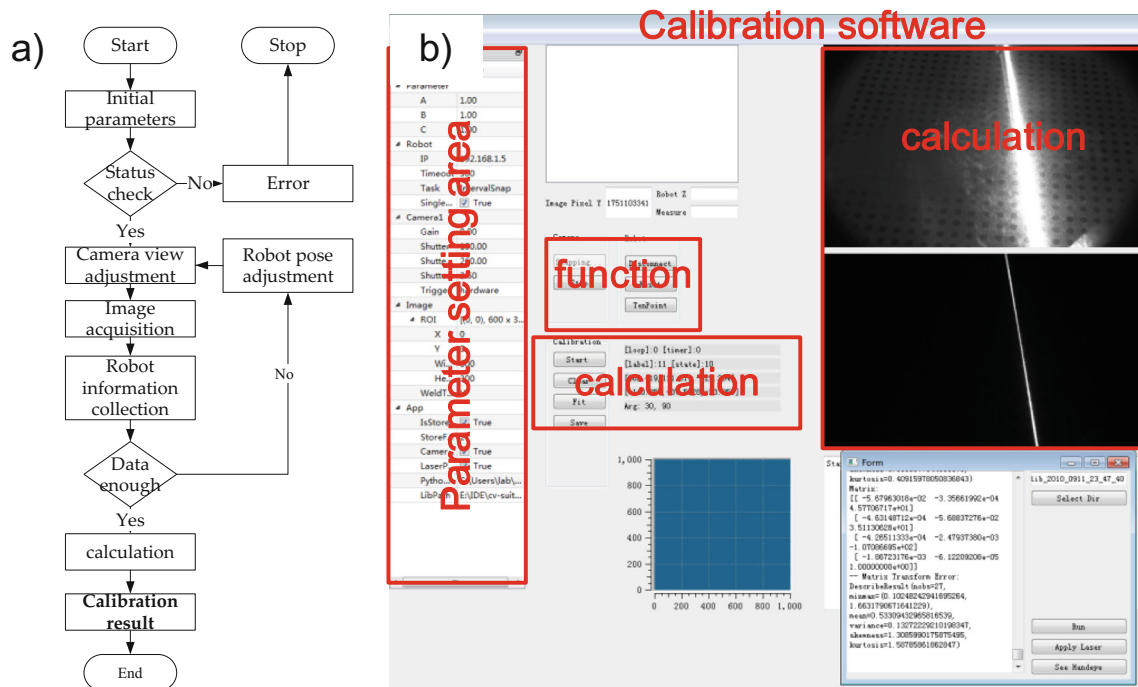


Fig. 8 Flow chart and interface of calibration software

to robot world coordinate form $M = (x, y, z)_{\text{world}}$ through the sensor frame.

Figure 11 shows the images captured during calibration and Table 2 shows the calibration results.

The result of calibration is verified by a single-point test: set a specific point that can be distinguished by the laser stripe, assign the extracted position to the robot position register, such as PR [1]. Accurately, move the robot to the specific point and record the position to another position register, such as PR [2]. The calibration result is evaluated by the deviation between the two position

registers. Figure 12 shows the results of the single-point test with the maximum error of less than 0.1 mm.

2.4 Image acquisition and processing

The arc intensity is much higher than the laser stripe emitted from the structured light vision sensor, according to the spectrum of Q235 GMAW process shown in Fig. 13. Specific wavelength of laser and narrow filter glass was chosen. Figure 14 shows the typical images acquired in fillet welding and V-groove welding.

There are two key points of image processing of active vision. One is the accuracy of the feature point extraction, especially for offline applications that usually require sub-pixel level accuracy. The other is the speed of image processing. In some applications of real-time processing, the

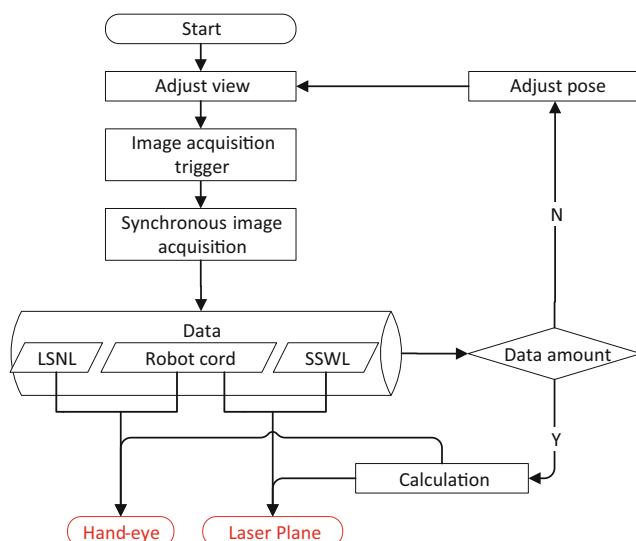


Fig. 9 Flowchart of fast-united calibration method

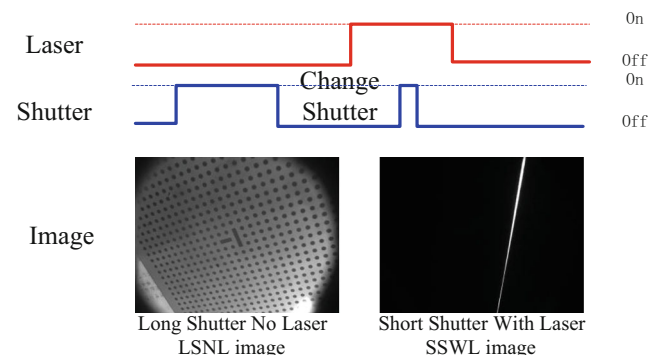
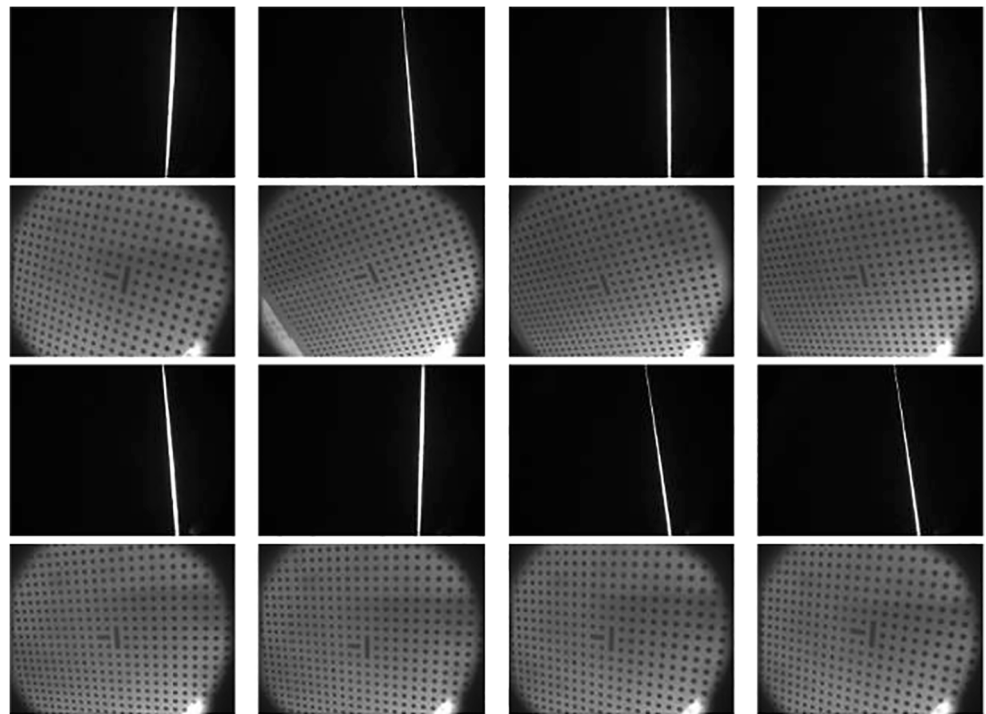


Fig. 10 Synchronous image acquisition process

Fig. 11 Images acquired for calibration



processing time of a frame of image is usually completed in 30 ms or less, depending on the length of the control cycle and the response speed requirements of the real-time processing.

Figure 15 shows the real-time seam tracking image processing method [31], which can extract feature points within 8 ms, but the extraction precision of feature points is 3~5 pixels, and the robustness under strong interference is not strong enough.

Figure 16 shows an offline processing flowchart based on a study of an improved Hough-processed laser image by Wu et al. [32]. Sub-pixel-level precision feature points can be extracted, but it takes more than 500 ms, which is not enough to meet real-time requirements.

The real-time requirements of the teaching-free intelligent welding image processing are very high, and the robustness and precision of the algorithm need to be guaranteed under strong interference. In this paper, an improved image

processing method based on a prior model and Steger method is proposed for laser vision image processing in GMAW process. The algorithm has excellent performance in terms of feature extraction precision, real-time performance, and robustness under strong interference.

The Steger method [33] uses the Hessian matrix for image convolution to obtain the line feature properties of each point in the image. This attribute was first widely used in medical images processing [34, 35], and then in machine vision applications [36, 37]. Like the Hough transform, the large amount of computation is the key reason why it is difficult to meet real-time requirements.

The improved image processing algorithm is divided into five steps: (1) classification based on a prior model; (2) ROI division; (3) line feature filtering based on Steger method; (4) sub-region division and angle filtering; and (5) feature point extraction.

Table 2 Calibration result of visual system

Parameter	Expression	Calibration result
Laser plane	$[A, B, C, D]$	$[-3281.414 \quad -66.789 \quad -491.162 \quad 99944.918]$
Hand-eye relationship	$H = \begin{bmatrix} R & T \\ 0^T & 1 \end{bmatrix}$	$\begin{bmatrix} 7.30 \times 10^{-1} & 1.87 \times 10^{-2} & -6.83 \times 10^{-1} & 132.30 \\ -1.75 \times 10^{-2} & 9.99 \times 10^{-1} & 8.66 \times 10^{-3} & -10.46 \\ 6.83 \times 10^{-1} & 5.64 \times 10^{-3} & 7.31 \times 10^{-1} & -38.91 \\ 0 & 0 & 0 & 1.00 \end{bmatrix}$
Sensor frame	$[x \ y \ z \ w \ p \ r]$	$[132.30 \quad -10.46 \quad -38.91 \quad 0.44 \quad -43.07 \quad -1.37]$

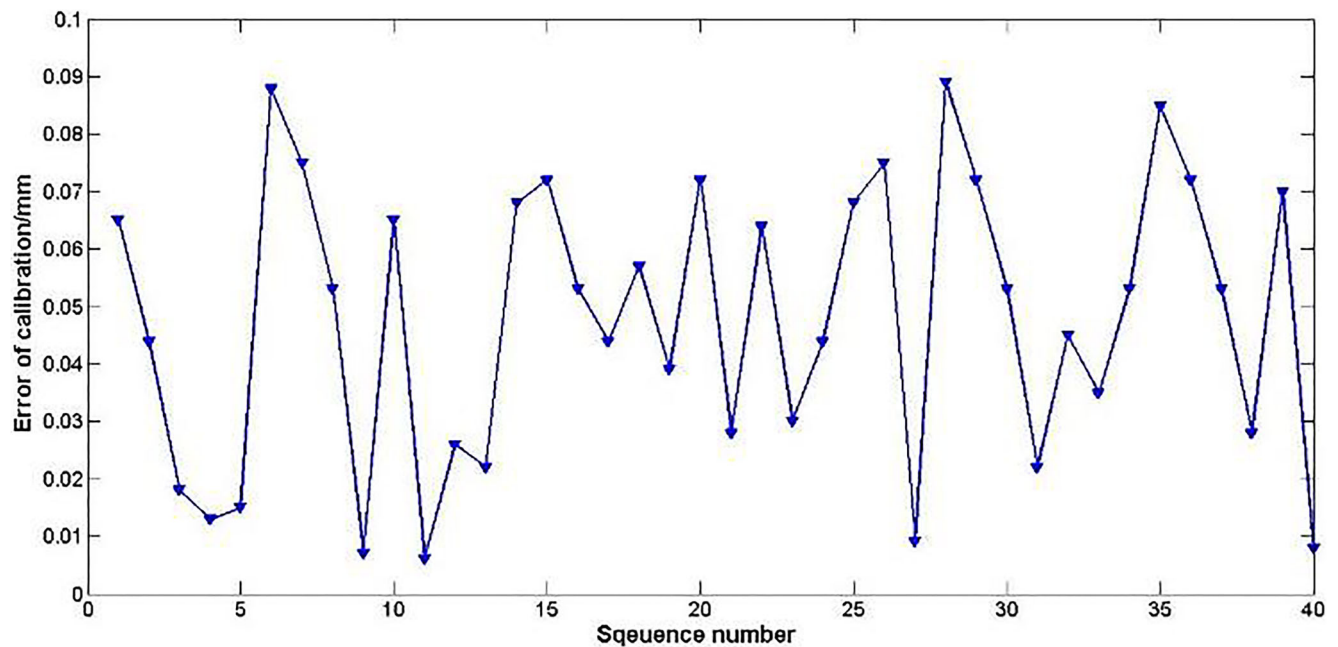


Fig. 12 Tests of calibration result by single point test

2.4.1 Classification based on a prior model

By observing the typical welds and their laser vision images, it can be seen that the laser vision images of typical welds can be simplified into a priori model consisting of points and lines, used to describe the structural characteristics of typical welding grooves, as shown in Fig. 17.

The priori models of the above four typical weld grooves correspond to different numbers of slope mutation points, respectively. According to this, the first image captured before the welding without welding interference can be simply processed by the threshold and centroid method. Laser stripes and slope mutation points on the stripes can be easily extracted, and the welding groove is classified as the type of the priori model.

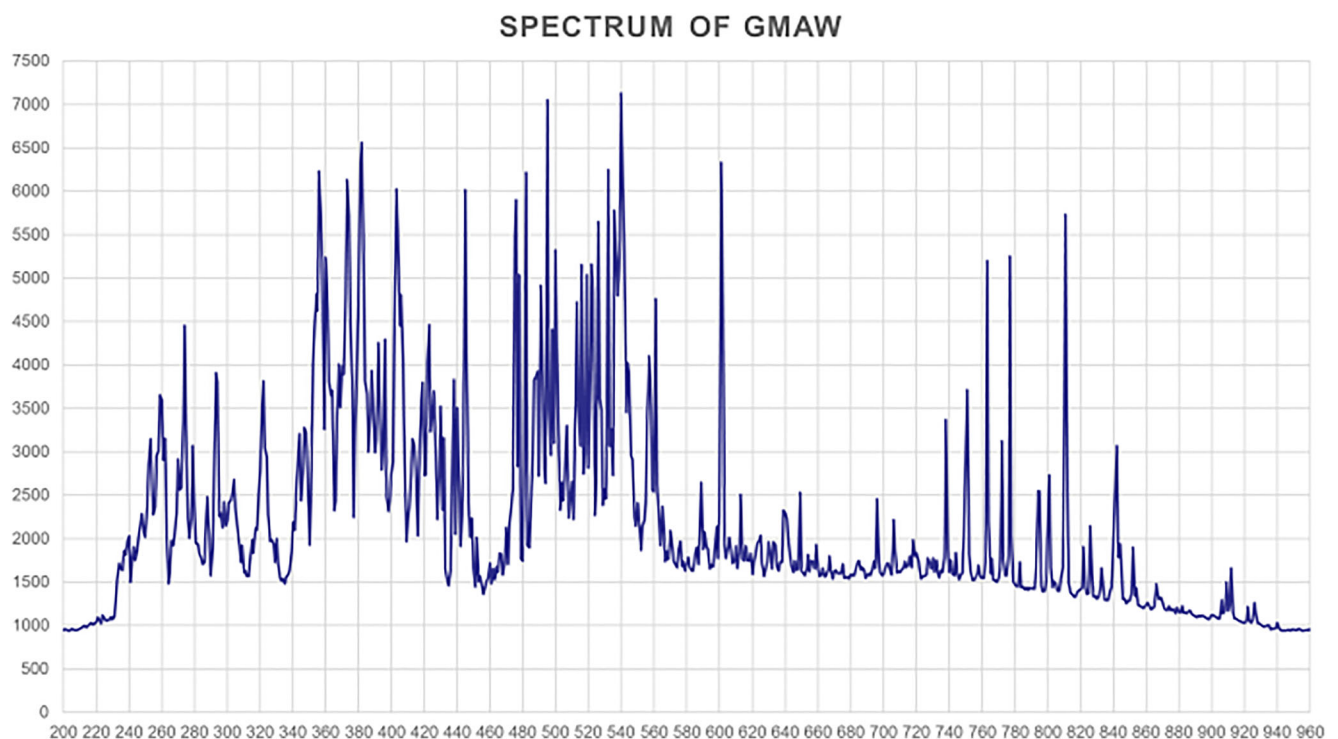


Fig. 13 Spectrum of Q235 GMAW

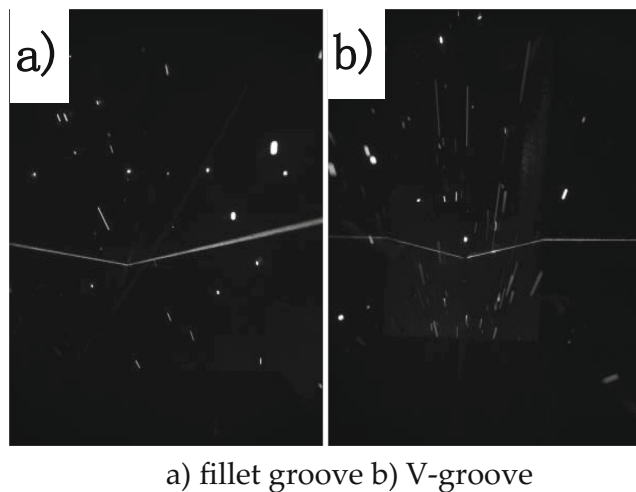


Fig. 14 Acquired image during welding. **a** Fillet groove and **b** V-groove

2.4.2 ROI division

The laser stripes occupied a small area in the entire image. Extracting the region of interest (ROI) where the laser stripes are located can greatly improve image processing speed and eliminate the interference in non-welded area. In a separate welding process, the relative position of the sensor and the workpiece is maintained, and the position of the welding area appearing in the image will also remain constant, so selecting a fixed ROI is the best choice. The ROI division can be expressed as formula (5):

$$r = [u \quad v \quad W \quad H] \quad (5)$$

where (u, v) is the center of ROI in image coordinate system, and W and H represent width and height of ROI respectively.

The initial image is processed to determine the initial value of r . Generally, the first image containing the characteristic laser stripes formed on the workpiece is obtained from a safe point. The distance between the workpiece along the axis of the vision sensor and the camera is defined as the weld height. The welding height of the safe point is significantly higher

than in the actual welding condition. The height of the weld determines the vertical coordinates of the laser stripes that appear in the image. Therefore, there is a significant difference between the vertical coordinate of the stripe in the image obtained at the safe point and the actual welding position, but we can use the laser plane function and the hand-eye relationship matrix to convert the image features into robot-executable coordinates and move the robot to reach that point. At this time, it has the same welding height as the actual welding. According to its feature point position (u_f, v_f) , the initial value of r can be expressed as formula (6).

$$\begin{cases} (u, v) = (u_f, v_f) \\ W = \frac{1}{2}(u_{\max} - u_{\min}) \\ H = v_{\max} - v_{\min} + \alpha\Psi \end{cases} \quad (6)$$

where (u_f, v_f) is the feature point position in image coordinate; $u_{\max}$, $u_{\min}$, $v_{\max}$, and $v_{\min}$ are the maximum and minimum of u and v axis of extracted laser stripe in the image coordinate, α is the proportional factor of laser stripe, and Ψ is the width of laser stripe.

2.4.3 Line feature filtering based on Steger method

As shown in Fig. 18, if the 2-D image is extended to the 3-D space with the gray value as the third axis, the linear feature of the laser stripe appears like ridge-like features, and at each point with line feature, the gradient along the line feature direction is extremely small, but the gradient perpendicular to the direction of line feature is extremely large.

In the classic Steger method, by calculating the Hessian matrix of each point in the image, the candidate points with line features in the image can be obtained. Hessian matrix of each pixel is calculated by formula (7).

$$H(x, y) = \begin{bmatrix} L_{xx} & L_{xy} \\ L_{xy} & L_{yy} \end{bmatrix} \quad (7)$$

Fig. 15 Fast image processing algorithm for real-time applications

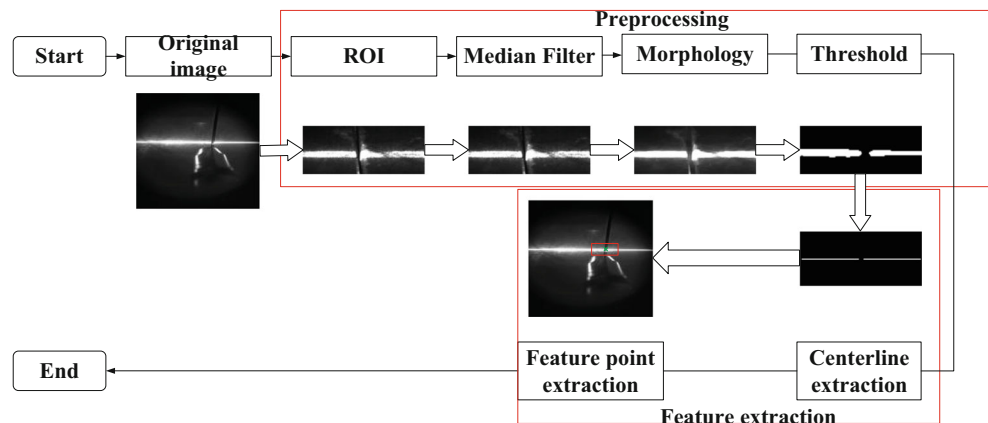
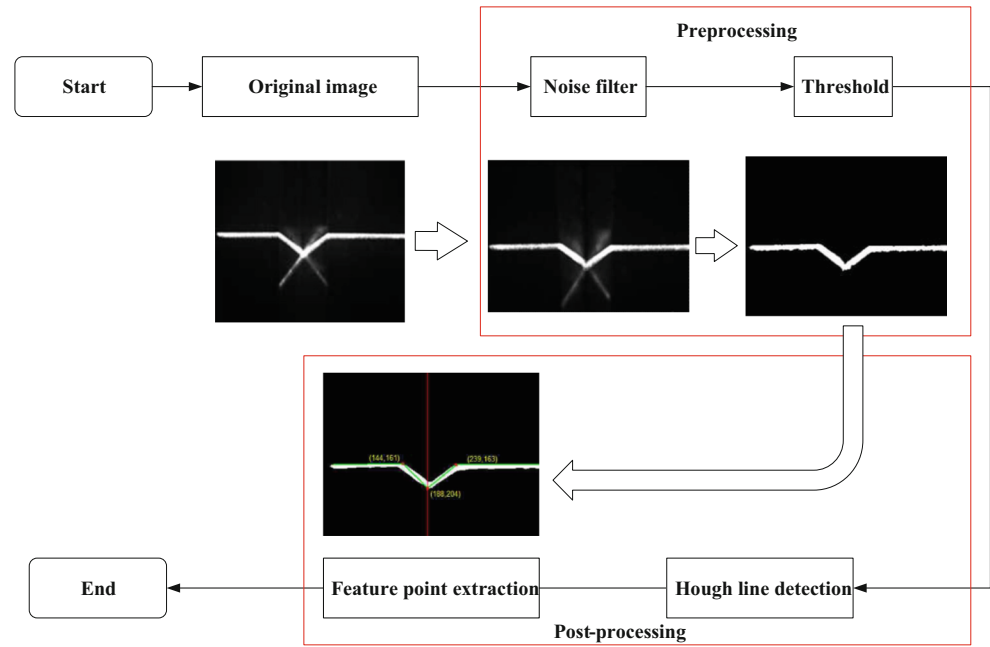


Fig. 16 Accurate image processing algorithm for offline applications



where L_{xy} is the derivative of the grayscale value in the x and y direction in order. L_{xx} and L_{yy} represent the second-order partial derivative of the grayscale value in the x and y direction respectively.

A grayscale image is a discrete 2-D integer matrix, which is the derivative of Gaussian function, and is used as the kernel to perform convolution operations to approximate the common derivation operations. Using I to represent a 2-D grayscale image, the convolution operation can be defined as formula (8).

$$L(x, y; \sigma) = g(x, y; \sigma) * I \quad (8)$$

where $*$ represents convolution operation, $g(x, y; \sigma)$ is the convolution kernel, using the Gaussian function, as formula (9) shows, its first and second derivatives are shown in formulas (10) and (11).

$$g(x, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}} \quad (9)$$

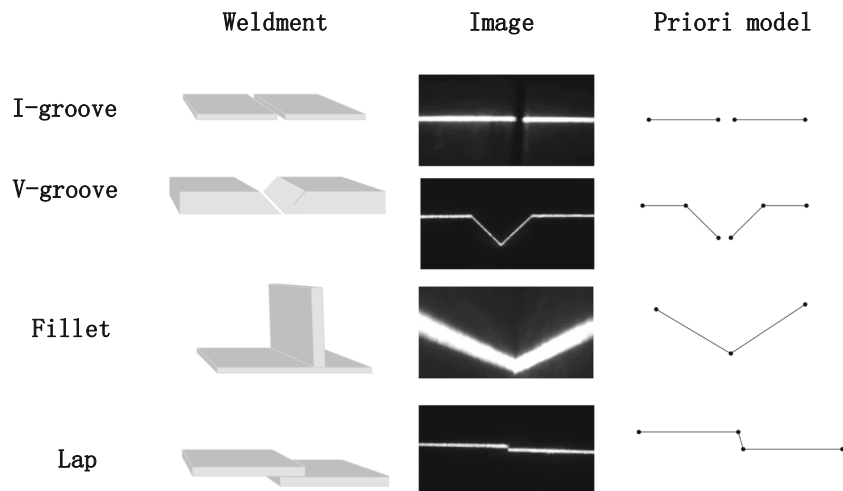
$$g'(x, \sigma) = \frac{-x}{\sqrt{2\pi}\sigma^3} e^{-\frac{x^2}{2\sigma^2}} \quad (10)$$

$$g''(x, \sigma) = \frac{x^2 - \sigma^2}{\sqrt{2\pi}\sigma^5} e^{-\frac{x^2}{2\sigma^2}} \quad (11)$$

where σ is the standard deviation of the Gaussian function determining the magnitude of the Gaussian kernel used for convolution, which is related to the line width to be detected.

As a real symmetric matrix, the eigenvectors of the Hessian matrix are orthogonal. It describes the second derivative of the gray value around the pixel (x_0, y_0) .

Fig. 17 Typical welding joint's laser image and its priori model



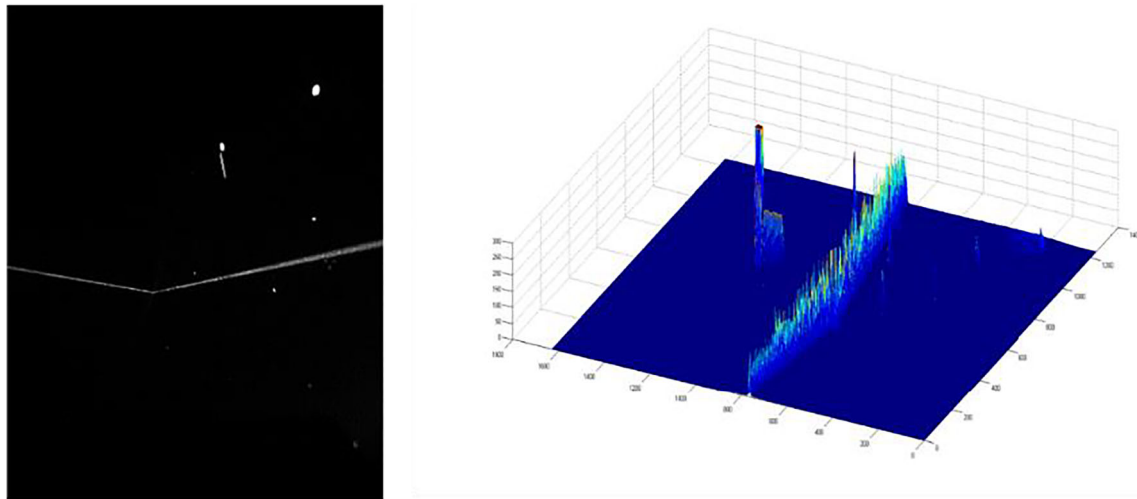


Fig. 18 2D and 3D perspective of laser stripe image

The Hessian matrix can determine the normal of the line feature. The feature vector corresponding to the maximum absolute value of the feature vector is the normal of the line to be detected, expressed as (n_x, n_y) . On this basis, through the normal Taylor expansion, the zero-crossing point (p_x, p_y) of the first derivative of the gray value in the line normal section around the benchmark point (x_0, y_0) is obtained with sub-pixel accuracy coordinate, expressed as formula (12):

$$(p_x, p_y) = (x_0 + tn_x, y_0 + tn_y) \quad (12)$$

where t is the distance between the benchmark point (x_0, y_0) and the zero-crossing point (p_x, p_y) and can be expressed as formula (13):

$$t = -\frac{r_x n_x + r_y n_y}{r_{xx} n_x^2 + 2r_{xy} n_x n_y + r_{yy} n_y^2} \quad (13)$$

Point (p_x, p_y) is a candidate of line feature points set if (tn_x, tn_y) satisfies formula (14).

$$(tn_x, tn_y) \in [-0.5, 0.5] \times [-0.5, 0.5] \quad (14)$$

Direction angle θ of the candidate points also can be calculated as formula (15):

$$\theta = \tan^{-1} \frac{n_y}{n_x} \quad (15)$$

Using the classic Steger method, the points most likely to have line features have been extracted as a candidate point set. There are two problems with this set of candidate points. One is interference from arc, dusts, splashes, and reflections. The other is that the amount of calculation does not meet the requirements of real-time applications. ROI division can greatly reduce the calculation amount; the anti-interference performance is still an urgent problem to be solved.

2.4.4 Sub-area division and angle filtering

The prior model can be used not only to classify welding types but also to simplify subsequent image processing. Previously, we conducted a classification of weld types based on the prior model, which obtained not only the weld type but also the number of slope mutation points m and its approximate horizontal position $x_j, j = 0, 1 \dots m$. Define segmentation point D according to x_j and boundary of ROI in x direction X_{r-} and X_{r+} as formula (16) shows. The average slope can also be calculated as formula (17), then the sub-area can be divided as formula (18). In each sub-area, the candidate points filtered by angle filter, most of the interference, can be eliminated.

$$D_n = \begin{cases} x_n, n = 1, 2 \dots m \\ x_r - \frac{1}{2}w, (X_{r-}), n = 0 \\ x_r + \frac{1}{2}w, (X_{r+}), n = m + 1 \end{cases} \quad (16)$$

$$k_n = \frac{\sum_{x=x_{n-1}}^{x_n} k_x}{X_n - X_{n-1}} \quad (17)$$

$$S_n : (x_{n-1}, x_n), n = 1, 2 \dots, m + 1 \quad (18)$$

Considering the fluctuation of the slope of pixels near the mutation points, n_t pixels near the mutation point are removed from sub-area division. Formulas (17) and (18) have been modified to formulas (19) and (20). The inclination angle replaces slope to express the direction of the line feature as formula (21).

$$k_n = \frac{\sum_{x=x_{n-1}+t}^{x_n-t} k_x}{x_n - x_{n-1} - 2n_t} \quad (19)$$

$$S_n : (x_{n-1} + n_t, x_n - n_t), n = 1, 2 \dots, m + 1 \quad (20)$$

$$\theta = \tan^{-1}k \quad (21)$$

Filtering angle of candidate points in each sub-area as formula (22):

$$|\theta_p - \theta_m| \leq \theta_e \quad (22)$$

where θ_p is the inclination angle of line feature angle of the filtering point, θ_m is the average angle of sub-area m , and θ_e is the angle tolerance, and expressed as formula (23):

$$\theta_e = \tan^{-1}(2\Psi/W) \quad (23)$$

2.4.5 Feature extraction

Considering the width of laser stripe and irregular reflection at the groove, the candidate points in each sub-area can be divided as “strong point” and “weak point.” According to its distance from the “fitted line,” Random Sample Consensus (RANSAC) fitting is an ideal method to solve the line feature extraction.

The feature point representing the center of the groove can be extracted as intersection of the fitted lines extracted from two sub-areas in the middle.

Figure 19 shows the flowchart of the proposed image processing method, which could be completed within 20 ms to meet the requirements of real-time processing.

In Fig. 20, the proposed method applied on an image with strong welding interferences, the result shows that the robustness of the proposed method is very strong.

2.5 Welding trajectory online planning

As shown in Fig. 21, the entire welding process starts from the arrival of the laser to the arc-on point and ends when the torch reaches the arc-end point. The laser vision sensor is installed at the front of the welding torch. There is a front distance in this kind of installation, the front distance is defined as the distance between the laser stripe and the torch along the welding direction at a certain height. In the teaching-free intelligent welding process, the time for laser to reach the arc-on point and the arc-end point is recorded as t_0 and t_2 , and the time for the welding torch to reach the arc-on point and the arc-end point is recorded as t_1 and t_3 . Figure 22 shows the three stages of teaching-free intelligent welding among the four status points: the guiding stage, the arc-on stage, and the arc-end stage. The scanning of laser stripe and torch represents trajectory planning and trajectory execution. For the entire welding process, welding is planned, and the spatially planned trajectory point is ahead of the welding position. The same position is planned earlier than execution in time

Fig. 19 Flow chart of image processing of improved Steger method

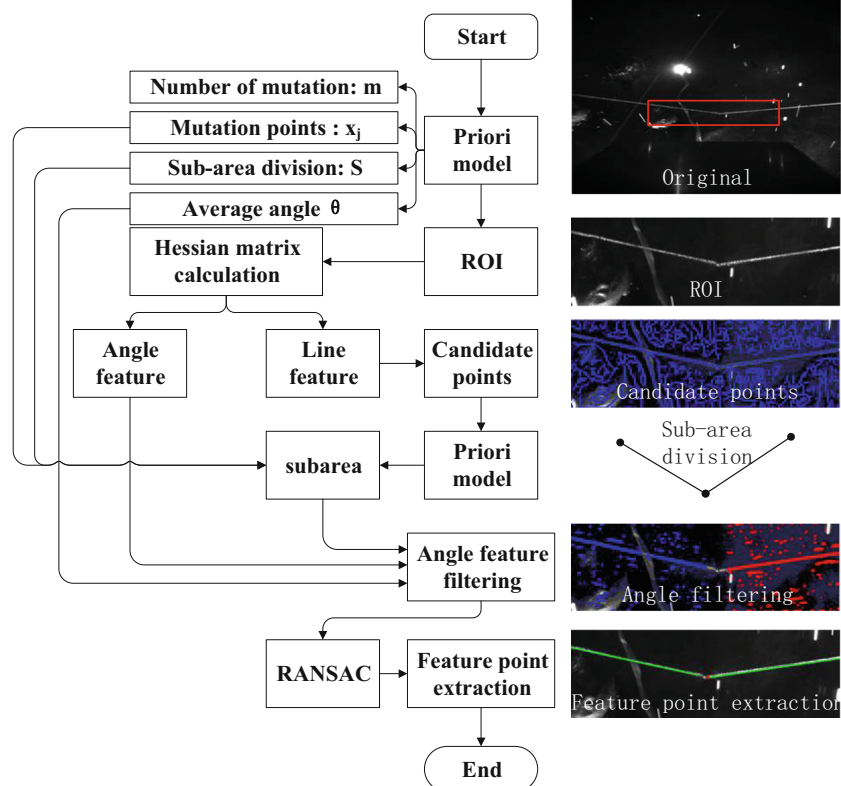
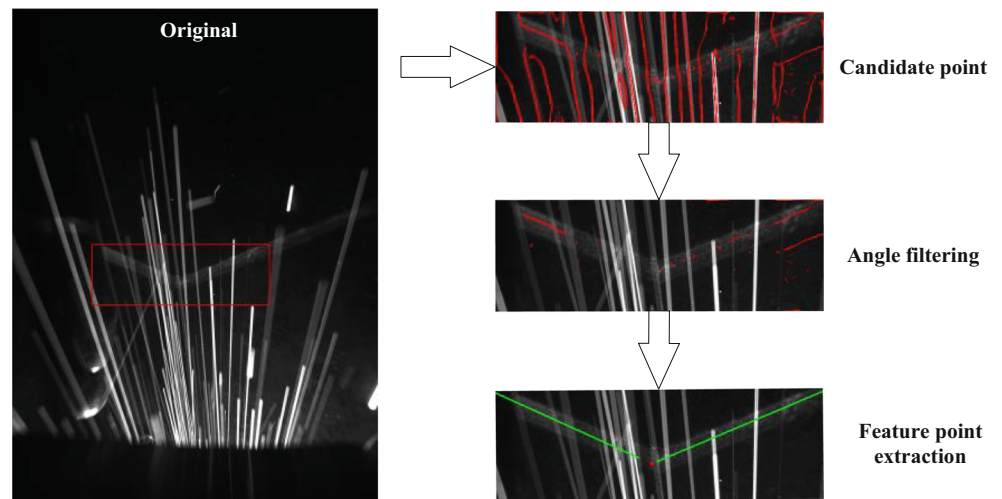


Fig. 20 Image process on image with strong interference

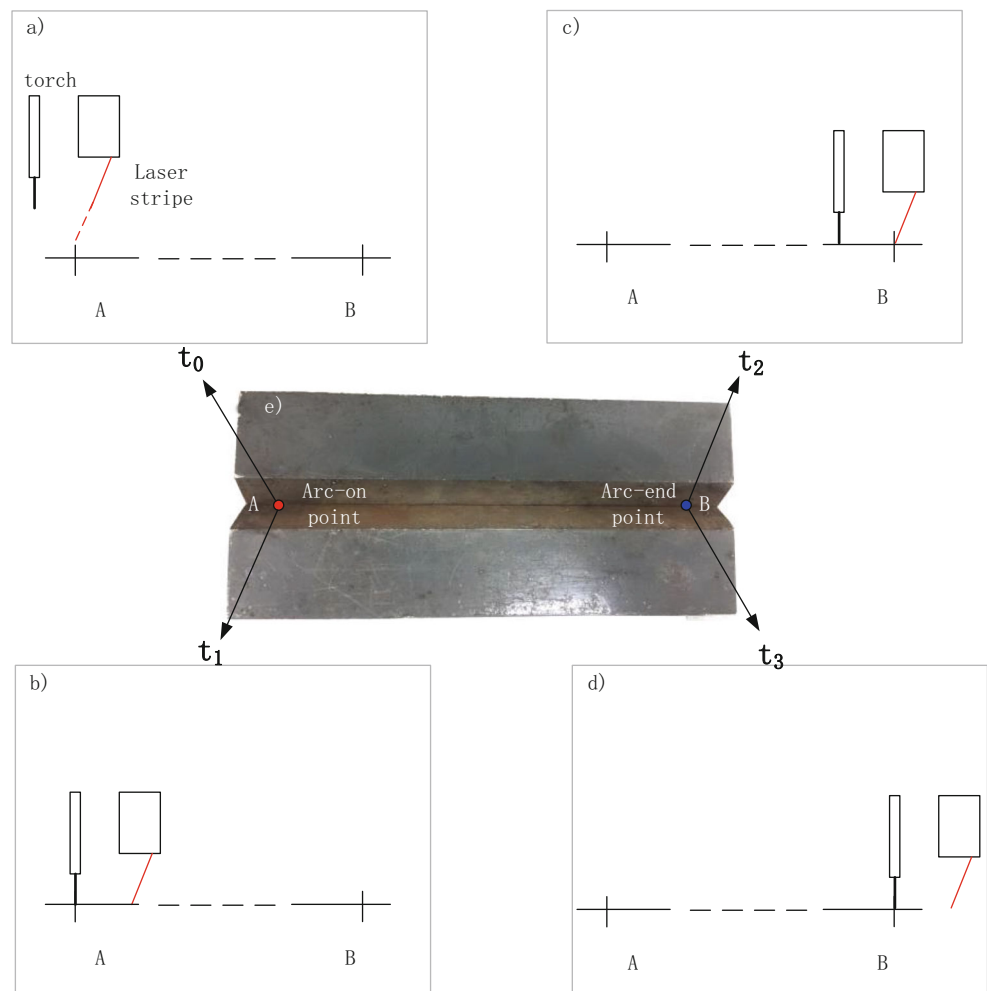


dimension. Therefore, it is necessary to store the planned but unexecuted trajectory points, the intermediate array which follows the first-in-first-out(FIFO) principle is suitable. At different stages, different operations are performed

on the intermediate array to implement the teaching-free intelligent welding process.

Figure 23 shows the operations of the intermediate array. Figure 24 shows the flow chart of the welding trajectory real-

Fig. 21 Torch and laser stripe status during welding process. **a–d** The schematic of position relation at time t_0 , t_1 , t_2 , and t_3



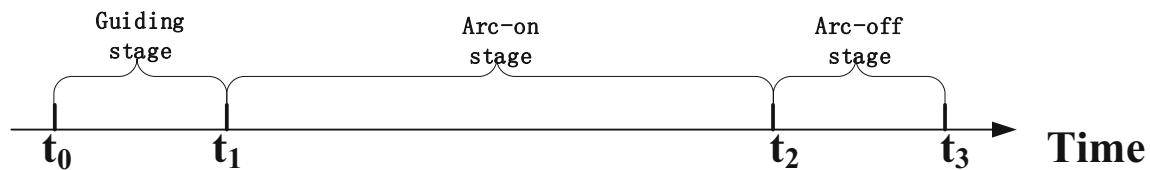


Fig. 22 Welding stage division

time planning process. Each stage will perform different operations. In the guiding stage, the laser vision sensor acquires the first image and calculates the first feature point. The intermediate array performs a save operation and a load operation, so that the robot starts to move toward to the first feature point position (the arc-on point). The continuous save operation performed during the movement. The criterion for the end of the guiding stage is the robot reaches the position of the first feature point (arc-on point).

When the robot reaches the arc-on point, it enters the arc-on stage. When the robot motion stops, the arc-on command will be executed. The intermediary array performs the load operation to start robot motion; the save and load operations will be continuously performed.

When cumulative planning trajectory L_p reaches the threshold value L_w , the arc-on stage ends. L_p can be calculated by formula (23).

Entering the arc-end stage, the laser vision sensor no longer collects images and provides new feature points. The

operation performed at this stage is load. The criterion for the end of this stage is that the intermediate array is empty.

$$L_p = \sum_{j=2}^m (T_j - T_{j-1}) + \sum_{i=2}^n (P_i - P_{i-1}) \quad (23)$$

3 Experiment and result

Experiment on V-groove weldment is carried out to testify the teaching-free welding method. The welding system used in the experiment is shown in Fig. 25. The weldment information and welding parameters are shown in Table 3. Figure 26 shows the original, welded part, and schematic.

A fine teaching was done manually before welding; the start-point and end-point were taught fine and accurately. Run the robot TP program to move the torch along the teaching trajectory without welding. During the process, the

Fig. 23 Structure and operations of intermediate array P

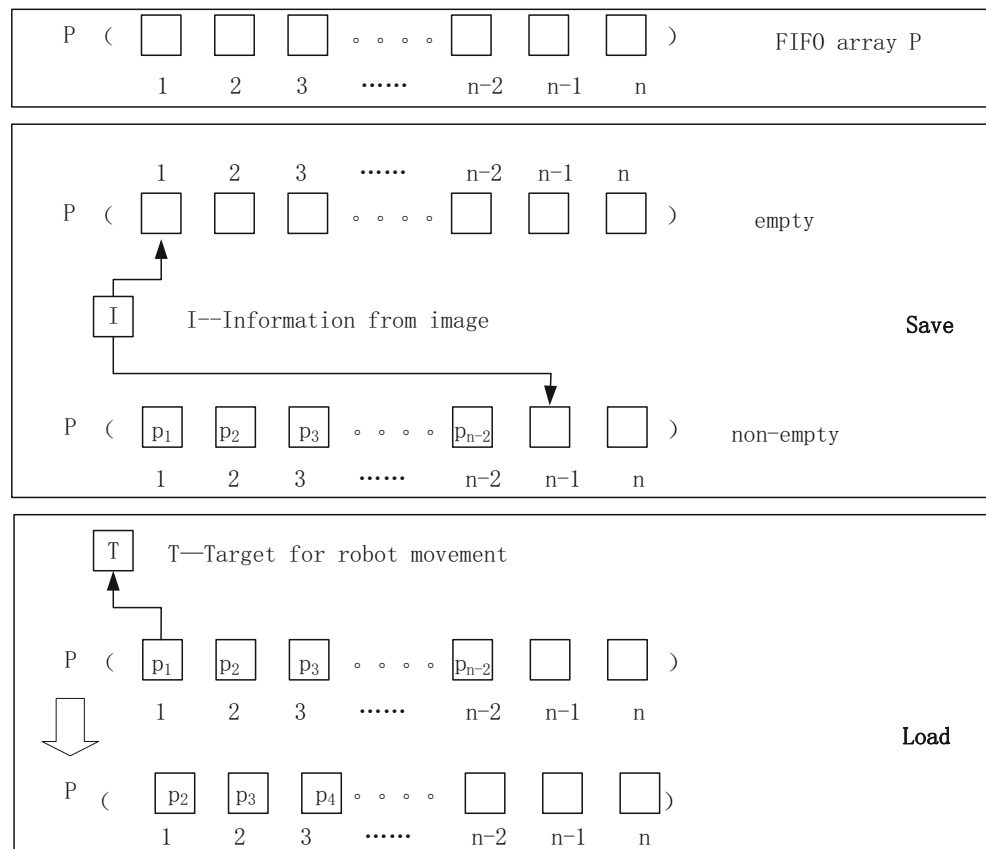


Fig. 24 Flow chart of welding trajectory online planning process

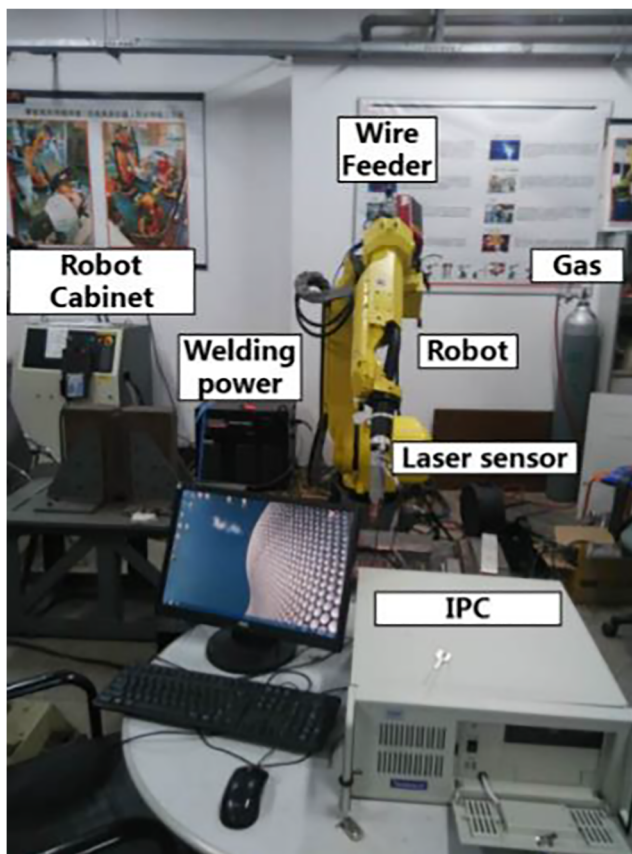
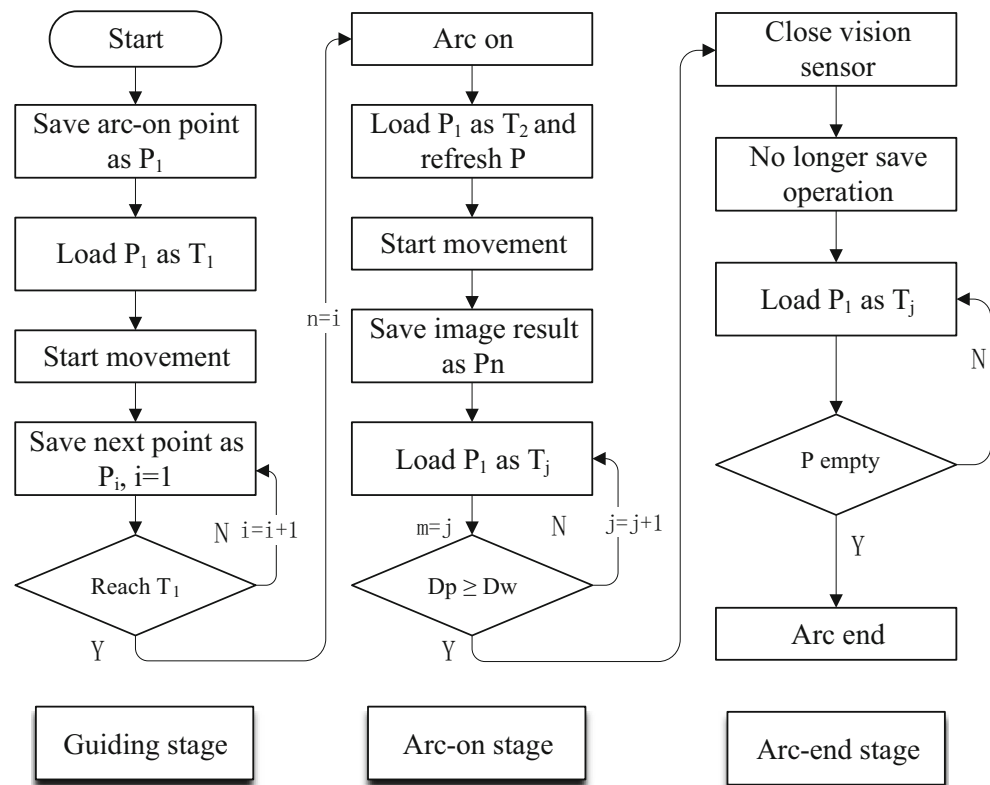


Fig. 25 Intelligent robotic welding system

trajectory robot covered was recorded as teaching trajectory. Open the real-time software and start TP program again with arc-on enable. Robot moves again according to the trajectory laser vision sensor planed in real-time; the trajectory robot covered was recorded as teaching-free trajectory.

Figure 27 shows the comparison of teaching trajectory and the teaching-free trajectory, which seems very close on the large axis scale. Take the corresponding points on the two trajectories along the Y axis for comparison.

Its error in X and Z axis shows in Fig. 28 and Fig. 29 respectively. The maximum error in X and Z axis is 0.19 mm and 0.10 mm respectively.

For further verification, the same experiment was done on fillet weldment. Weldment is shown in Fig. 30.

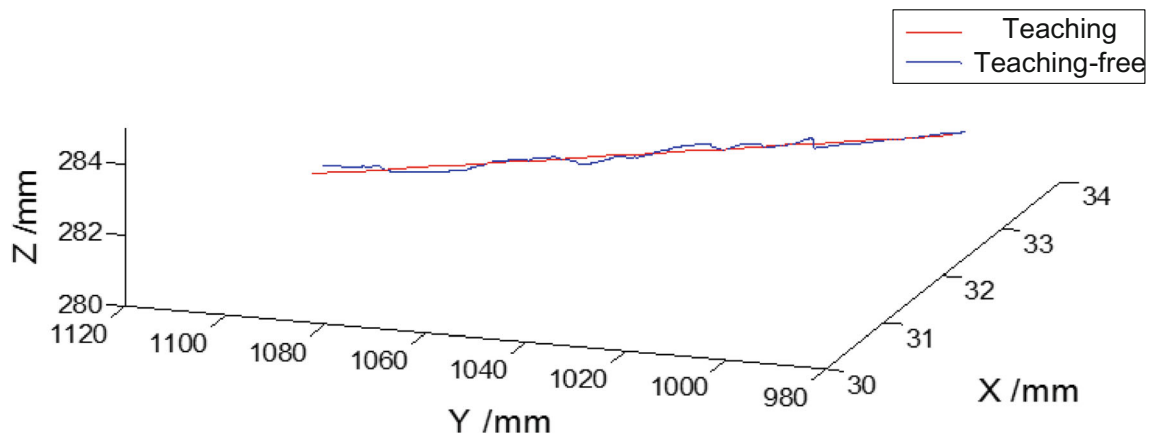
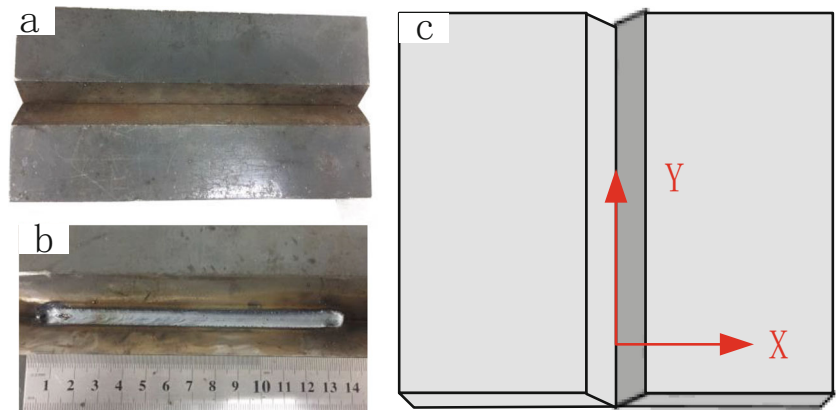
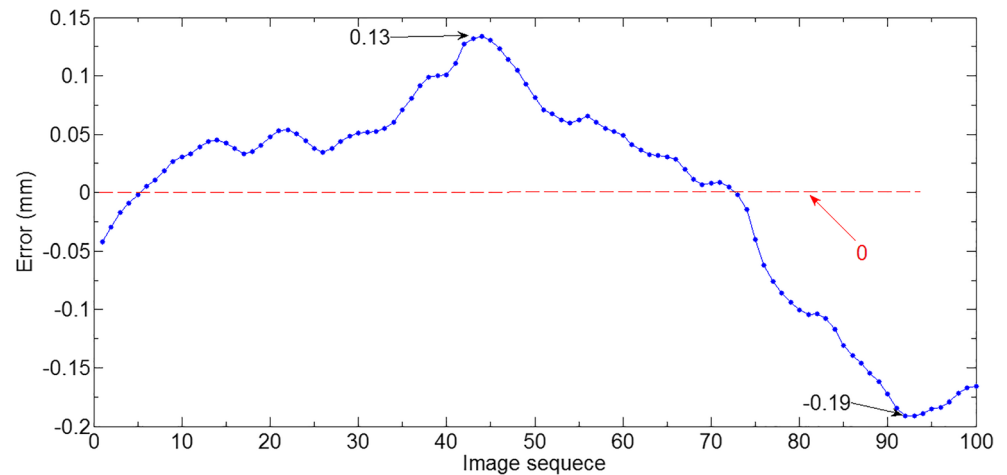
Result is shown in Figs. 31, 32, and 33. Result of fillet welding experiment shows its maximum error in X axis and Z axis is 0.21 mm and 0.08 mm respectively.

Table 4 shows the comparison with reports from other researchers [20, 21]; the teaching-free welding method satisfies the requirements of most welding applications in accuracy, robustness, and real-time performance.

The results of experiment concluded that the teaching-free welding method proposed in this paper can plan the welding trajectory during welding in real-time; its accuracy and robustness are well enough for most welding applications.

Table 3 The relevant technological parameters of experiment for Q235 steel sheet GMAW

Parameter	Value	Parameter	Value
Welding materials type of welding seam	Q235 V-groove	Welding speed	7 mm/s
		Wire diameter	1.2 mm
Pulse frequency	125 Hz	Shielding gas	80%Ar + 20%CO ₂
Welding current	150 A	Gas flow	15 L/min
Feed speed	5.0 m/min		

Fig. 26 V-groove weldment (**a** original weldment, **b** welded weldment, **c** schematic)**Fig. 27** Comparison of teaching trajectory and planning trajectory**Fig. 28** Error in X axis of teaching-free trajectory on V-groove welding

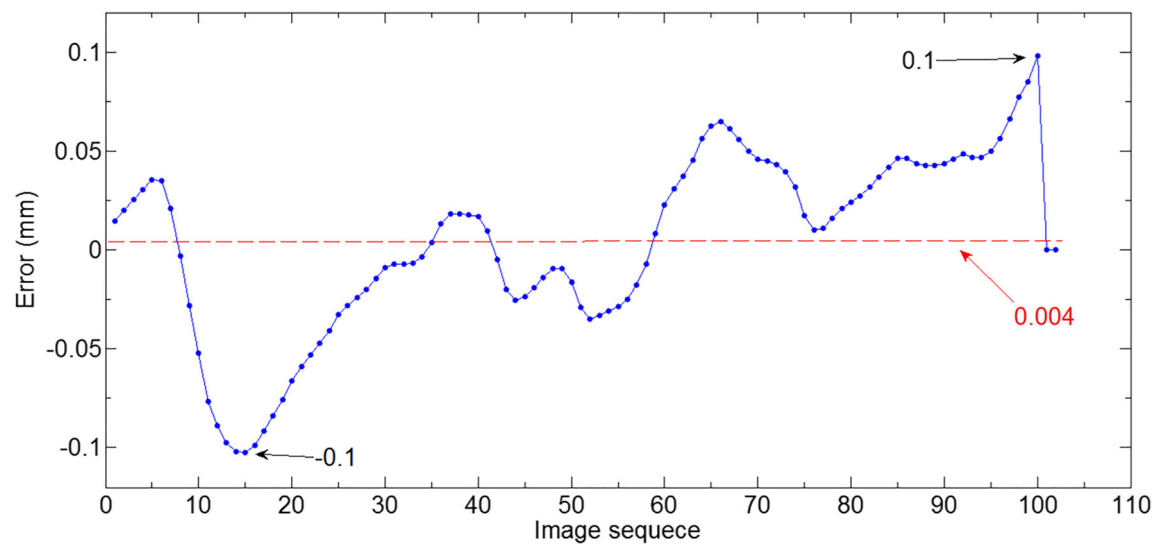


Fig. 29 Error in Z axis of teaching-free trajectory on V-groove welding

Fig. 30 Fillet weldment (**a** original weldment, **b** welded weldment, **c** schematic).

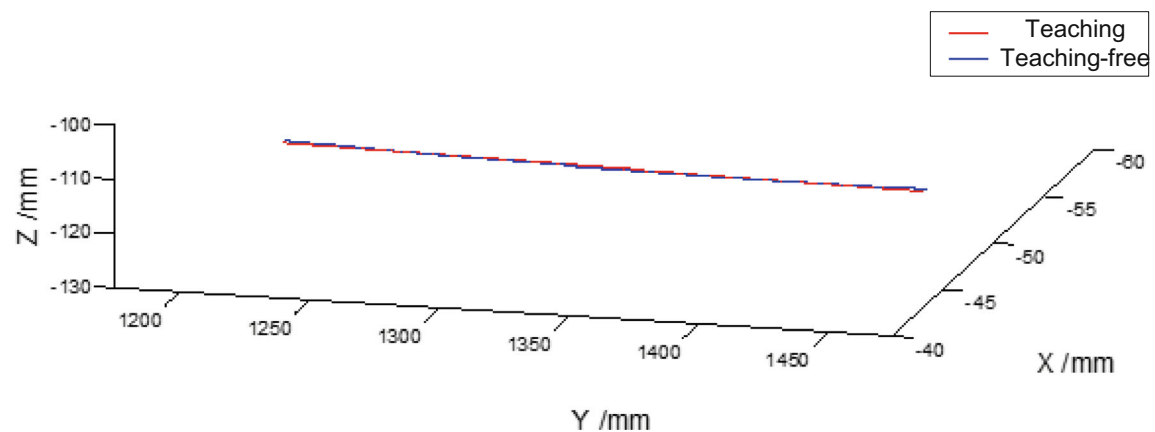
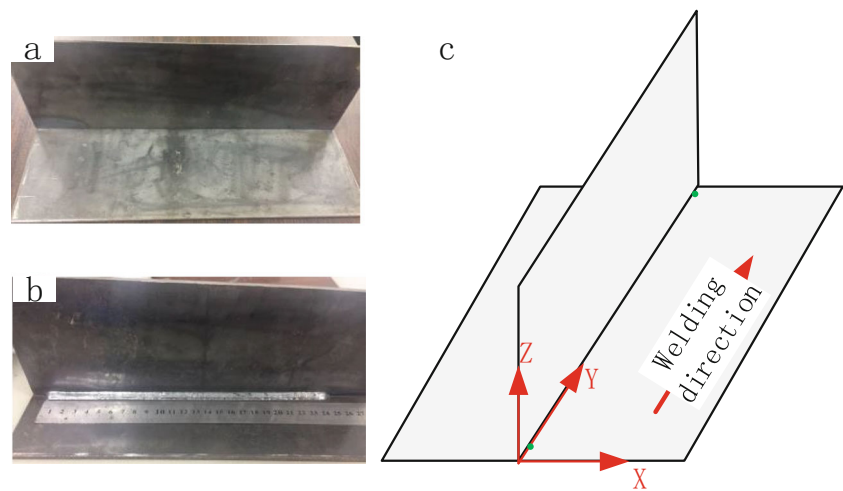


Fig. 31 Comparison of teaching trajectory and teaching-free trajectory on fillet welding

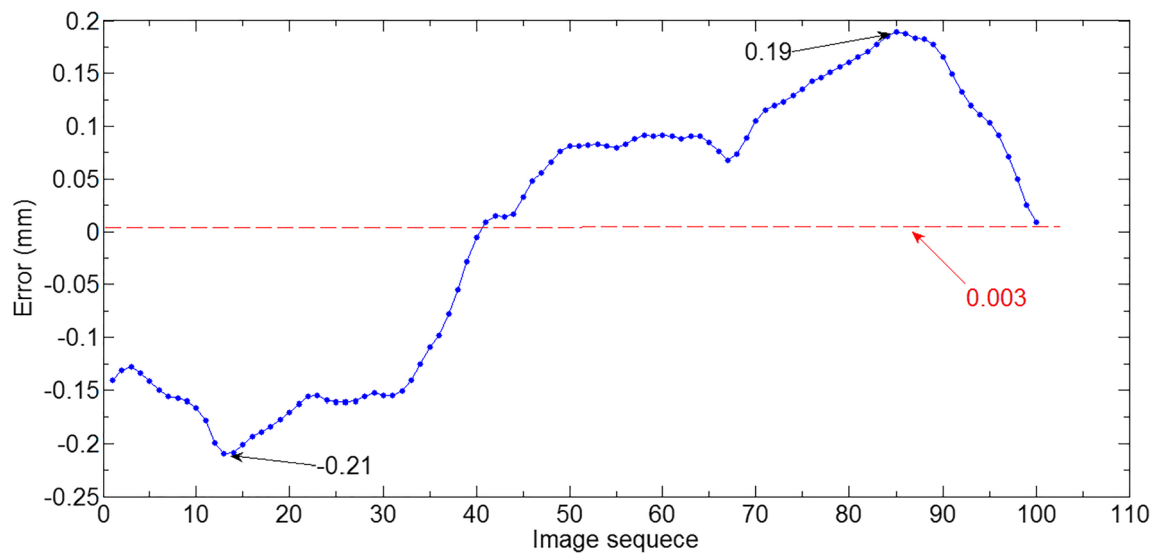


Fig. 32 Error in X axis of teaching-free trajectory on fillet welding

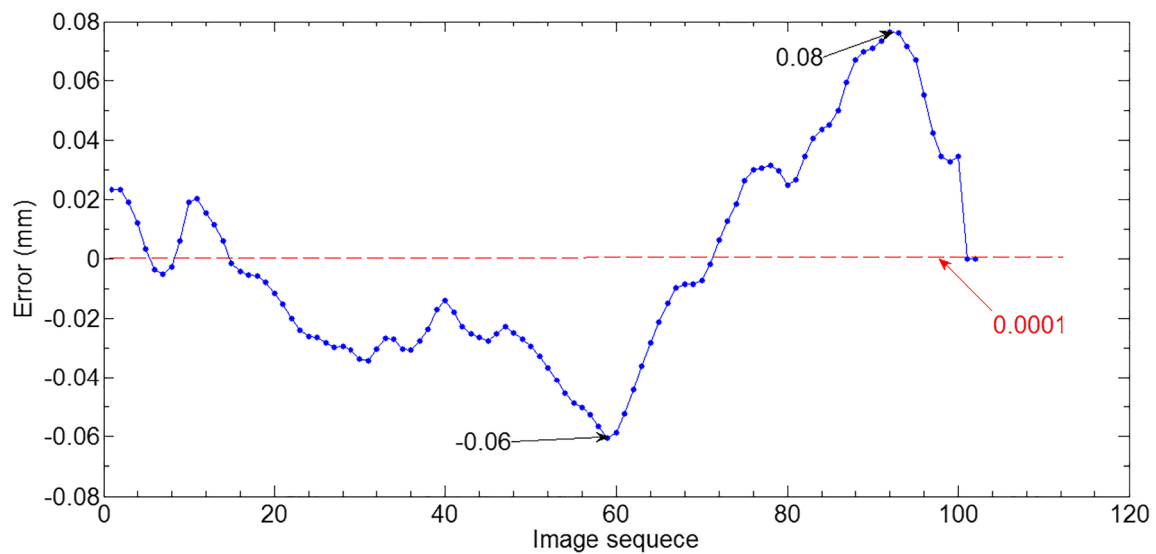


Fig. 33 Error in Z axis of teaching-free trajectory on fillet welding

4 Conclusions

Aiming at the problem that the trajectory needs to be taught in advance during the robot welding process in IRWT, this paper studies a teaching-free intelligent welding method based on laser vision. Concluded as follow:

1. A set of LVSS was established, including the development of laser vision sensor and software.
2. A fast-unity calibration method is proposed to improve the efficiency of LVSS calibration based on synchronous image acquisition. The calibration process could be done within 5 min.
3. A improved image processing algorithm based on a priori model and Steger method was proposed. The welding image features were extracted within 20 ms, with sub-pixel accuracy. Compared with the traditional laser vision image processing algorithms, the proposed algorithm has greatly improved the speed, accuracy, and robustness.
4. Based on the FIFO array communication, an online planning model for welding trajectory was proposed to realize teaching-free robot intelligent welding.

Table 4 Comparison with other method

Method	Accuracy (mm)
Teaching-free circular laser 3D [20]	0.2 1.0
Triple line [23]	0.5
CCOT [24]	0.2

5. In order to verify the accuracy and robustness of this method, the experiments of V-groove and fillet welds were conducted. The results showed that control accuracy is suitable for most robotic welding applications.

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References

1. Chen S (2015) On Intelligentized welding manufacturing, the advances in intelligent systems and computing, vol 363. Springer Verlag, pp 3–34
2. Chen S, Wu J (2009) Intelligentized methodology for arc welding dynamical processes[M] Springer-Verlag Berlin Heidelberg, Germany, Lecture Notes in Electrical and Engineering, LNEE 29
3. Chen S, Lv N (2014) Research evolution on intelligentized technologies for arc welding process. *J Manuf Process* 16:109–122
4. Chen S (2011) Research evolution on intelligentized technologies for robotic welding at SJTU. Lecture Notes in Electrical Engineering, Robotic Welding, Intelligence and Automation, LNEE v 88:3–14
5. Shen H, Wu J, Lin T, Chen S (2008) Arc welding robot system with seam tracking and weld pool control based on passive vision. *Int J Adv Manuf Technol* 39(7–8):669–678
6. Chen S (2007) On the key intelligentized technologies of welding robot, Lecture Notes in Control and Information Sciences, LNCIS 362, pp.105–116
7. Larkin N, Short A, Pan Z, van Duin S (2018) Automated programming for robotic welding. *Trans Intell Welding Manuf*:48–59
8. Erdős G, Kardos C, Kemény Z, Kovács A, Váncza J (2016) Process planning and offline programming for robotic remote laser welding systems. *Int J Comput Integr Manuf* 29(12):1287–1306
9. Zhang T, Ouyang F (2012) Offline motion planning and simulation of two-robot welding coordination. *Front Mech Eng* 7(1):81–92
10. Liu Y, Shi L, Tian X (2018) Weld seam fitting and welding torch trajectory planning based on NURBS in intersecting curve welding. *Int J Adv Manuf Technol* 95(5–8):2457–2471
11. Jeong SK, Lee GY, Lee WK, Kim SB (2001) Development of high speed rotating arc sensor and seam tracking controller for welding robots. *IEEE ISIE* 2001 2:845–850
12. Shi YH, Yoo WS, Na SJ (2006) Mathematical modelling of rotational arc sensor in GMAW and its applications to seam tracking and endpoint detection. *Sci Technol Weld Join* 11(6):723–730
13. Wang J, Chen B, Chen H, Chen S (2009) Analysis of arc sound characteristics for gas tungsten argon welding. *Sens Rev* 29(3):240–249
14. Song S, Chen H, Lin T, Wu D, Chen S (2016) Penetration state recognition based on the double-sound-sources characteristic of VPPAW and hidden Markov model. *J Mater Process Technol* 234:33–44
15. Huang Y, Wu D, Zhang Z, Chen H, Chen S (2017) EMD-based pulsed TIG welding process porosity defect detection and defect diagnosis using GA-SVM. *J Mater Process Technol* 239:92–102
16. Huang Y, Wu D, Lv N, Chen H, Chen S (2017) Investigation of porosity in pulsed GTAW of aluminum alloys based on spectral and X-ray image analyses. *J Mater Process Technol* 243:365–373
17. Shao W, Liu X, Wu Z (2019) A robust weld seam detection method based on particle filter for laser welding by using a passive vision sensor. *Int J Adv Manuf Technol* 104(5–8):2971–2980
18. Nilsen M, Sikström F, Christiansson AK, Ancona A (2019) Robust vision-based joint tracking for laser welding of curved closed-square-butt joints. *Int J Adv Manuf Technol* 101(5–8):1967–1978
19. Xu Y, Yu H, Zhong J, Lin T, Chen S (2012) Real-time image capturing and processing of seam and pool during robotic welding process. *Ind Rob Int J* 39.5(2012):513–523
20. Xu Y, Yu H, Zhong J, Lin T, Chen S (2012) Real-time seam tracking control technology during welding robot GTAW process based on passive vision sensor. *J Mater Process Technol* 212(8):1654–1662
21. Shen H, Ma H, Lin T, Chen S (2007) Research on weld pool control of welding robot with computer vision. *Ind Rob Int J* 34(6):467–475
22. He Y, Yu Z, Li J, Ma G (2019) Weld seam profile extraction using top-down visual attention and fault detection and diagnosis via EWMA for the stable robotic welding process. *Int J Adv Manuf Technol* 104(9–12):3883–3897
23. Du R, Xu Y, Hou Z, Shu J, Chen S (2019) Strong noise image processing for vision-based seam tracking in robotic gas metal arc welding. *Int J Adv Manuf Technol* 101(5–8):2135–2149
24. Xu P, Xu G, Tang X, Yao S (2008) A visual seam tracking system for robotic arc welding. *Int J Adv Manuf Technol* 37(1–2):70–75
25. Luo H, Chen X (2005) Laser visual sensing for seam tracking in robotic arc welding of titanium alloys. *Int J Adv Manuf Technol* 26(9–10):1012–1017
26. Fang Z, Xu D, Tan M (2011) A vision-based self-tuning fuzzy controller for fillet weld seam tracking. *IEEE/ASME Trans mechatron* 16(3):540–550
27. Zou Y, Chen X, Gong G, Li J (2018) A seam tracking system based on a laser vision sensor. *Measurement* 127:489–500
28. Zou Y, Chen T (2018) Laser vision seam tracking system based on image processing and continuous convolution operator tracker. *Opt Laser Eng* 105:141–149
29. Xu Y, Fang G, Chen S, Zou J, Ye Z (2014) Real-time image processing for vision-based weld seam tracking in robotic GMAW. *Int J Adv Manuf Technol* 73(9–12):1413–1425
30. Zhang Z (2000) A flexible new technique for camera calibration. *IEEE Trans Pattern Anal Machine Intell* 22
31. Huang S (2017) Research on typical weld seam feature extraction and real-time seam tracking based on active vision M.S. thesis, Department of material science and engineering, Shanghai Jiao Tong University
32. Wu Q, Lee J, Park M, Jin B, Kim D, Park C, Kim I (2015) A study on the modified Hough algorithm for image processing in weld seam tracking. *J Mech Sci Technol* 29(11):4859–4865
33. Steger C (1998) An unbiased detector of curvilinear structures. *IEEE Trans Pattern Anal Machine Intell* 20(2):113–125
34. Zwiggelaar R, Astley S, Boggis C, Taylor C (2004) Linear structures in mammographic images: detection and classification. *IEEE Trans Med Imag* 23(9):1077–1086
35. Matsopoulos G, Asvestas P, Mouravliansky N, Delibasis K (2004) Multimodal registration of retinal images using self organizing maps. *IEEE Trans Med Imag* 23(12):1557–1563
36. Navab N, Genc Y, Appel M (2000) Lines in one orthographic and two perspective views. *IEEE Conf CVPR* 2000 (Cat. No. PR00662) 2. IEEE
37. Sun X, Gao G, Tian B et al (2019) Intelligent recognition of production date based on machine vision. *Journal of Physics: Conference Series*. IOP Publishing 1267(1):012041

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